LEARNER GUIDE

Expressions, equations, formulae and identities

Algebraic Expressions and Identities · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

An **expression** is a value, such as $3x + 4$. An **equation** is true only for particular values: $3x + 2 = 14$ is true only when $x = 4$. A **formula** links quantities that vary, such as $A = \pi r^2$. An **identity** is true for every value, and is written with $\equiv$: $2(x + 3) \equiv 2x + 6$. Testing values can suggest an identity. One **counterexample** shows a claim is false. Only algebra, such as expanding, **proves** an identity.

What you are learning to notice

- For an identity, the graphs of the two sides are the same line. For an equation, they meet only where it is true.
- Agreeing at a few values proves nothing: $x + x$ and x^2 agree at 0 and 2, but they are not the same.
- One counterexample is enough: at $x = 1$, $(x + 1)^2 = 4$ but $x^2 + 1 = 2$.
- To prove an identity, expand and simplify one side until it matches the other.
- To find missing numbers in an identity, expand and match the x terms and the number terms.

Moving through it, one step at a time

Test a few values to get a feel. If two values disagree, you have a counterexample: it is not an identity. If they always agree, expand and simplify one side until it is exactly the other side: that is the proof. Write $\equiv$ once it is proved.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Test, then prove

Is $2(x + 3) = 2x + 6$ an identity?

1. Test $x = 1$: $2 \times 4 = 8$, and $2 + 6 = 8$.	they agree
2. Test $x = -2$: $2 \times 1 = 2$, and $-4 + 6 = 2$.	they agree
3. Testing suggests an identity, but cannot check every number.	so prove it
4. Expand: $2 \times x + 2 \times 3$.	$= 2x + 6$

$2(x + 3) \equiv 2x + 6$: proved for every value of x.

Example 2 — A counterexample

Is $(x + 1)^2 = x^2 + 1$ an identity?

1. Test $x = 0$: 1 and 1.	they agree
2. Test $x = 1$: $(1 + 1)^2 = 4$, but $1 + 1 = 2$.	they differ
3. One value where they differ is enough.	not an identity
4. Expanding shows why: $(x + 1)^2 = x^2 + 2x + 1$.	the $2x$ is missing

Not an identity. It is an equation, true only when $x = 0$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (substitution, including negative numbers, and expanding a bracket), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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